

Deepak Mallampati

RESEARCH ENGINEER - APPLIED ML & CLINICAL NLP

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/deepak-glitch |
US-based, F-1 OPT · honest re: on-site India relocation

PROFESSIONAL SUMMARY

Applied ML / AI Engineer with 4+ years building **clinical NLP and LLM systems over regulated healthcare data**. Delivered RAG pipelines at **92% precision** with **40%** hallucination reduction, BERT/roBERTa clinical NER at **89% F1**, and privacy-preserving clinical ML that cut re-identification risk **3.38% → 0.32%** while retaining 99% baseline performance. Strong in extraction logic, evaluation, error analysis, and fine-tuning (QLoRA on H100).

CORE COMPETENCIES & SKILLS

Clinical NLP · Information Extraction · LLMs & RAG · Fine-tuning (QLoRA) · Model Evaluation & Error Analysis · Transformers & Embeddings · Structured Outputs · Privacy-Preserving ML

ML & NLP: PyTorch, TensorFlow · HuggingFace Transformers · BERT/roBERTa NER · Text Classification, Summarization · precision/recall/F1 · Error Analysis · Experiment Tracking (MLflow, W&B)

LLMs & GenAI: GPT-4o, Claude, LLaMA, Qwen · RAG · Prompt Engineering · Fine-tuning (LoRA/QLoRA/PEFT) · Structured LLM Outputs · LangChain/LangGraph

Retrieval: FAISS, Pinecone, Weaviate · Embeddings · Cross-Encoder Re-ranking · Semantic Search

Clinical & Compliance: HIPAA · Clinical-text corpora · K-anonymity, L-diversity, Differential Privacy · PII Redaction

Engineering: Python · FastAPI · Docker, Kubernetes · APIs · Spark, Kafka data pipelines

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- **Built clinical information extraction + RAG over healthcare document corpora** - extraction logic, evaluation scripts, and error-analysis tooling for domain-specific clinical queries.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.
- Optimized inference cost/latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving clinical ML pipeline** on a hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.
- Engineered content-safety safeguards for policy-compliant AI responses, reducing unsafe outputs by **42%**.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.
- Built supervised regression/classification models for equipment-failure prediction and anomaly detection with end-to-end data preprocessing.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy on a real hospital-readmission dataset; cut re-identification risk 3.38% → 0.32% retaining 99% baseline performance. IEEE paper.

Story Director LLM

Fine-tuned (QLoRA) language model generating structured narratives from a single prompt, with an automated rendering pipeline.

career-ops

Autonomous AI-driven job-search pipeline built on Claude Code / agentic tooling - scan, score, and tailor applications overnight.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 · GPA 3.70/4.0